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FPGA-based ultrasonic energy mapping with source removal method for damage visualization in composite structures

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This paper presents the implementation of an ultrasonic energy mapping (UEM) method in a field programmable gate array (FPGA)-based processing board. This method is integrated into an ultrasonic propagation imaging (UPI) system for real-field application. UEM method involves two-dimensional Fourier transform and wavenumber filtering to capture the energy scattering waves. Performing multi-dimensional Fourier transform in FPGA is a challenging task because it cannot fit into a single FPGA chip, and therefore, external memory is required to handle a large amount of data. The row–column decomposition method is used to perform 2D Fourier transform using two DDR3 SDRAM memories. Additionally, a source removal function is added into the system to highlight the damage when high energy at the source location occurs. The UPI system was applied for inspection of a composite wing box with impact damages. The UEM results showed that a damage of $20 \times 20 \text{ mm}^2$ was detected successfully in the presence of the source inside the scan area. Moreover, the UEM result is produced in less than 5 s for 57,600 points, which is practical for a real-world, non-destructive testing application.

Keywords: non-destructive testing; ultrasonic propagation imager; ultrasonic energy mapping; composite structures; field programmable gate array

1. Introduction

Composite materials are widely used in military and aerospace applications because they offer weight reduction and greater strength, which is critical to achieving higher speed and increased payloads [1]. The non-destructive testing (NDT) is crucially important for composite structures during manufacturing, as well as in-service maintenance and repair. Since composite materials show complex fracture characteristics and mechanical behavior, it is difficult to perform NDT of composites by conventional technologies [2,3]. Many researchers have studied Lamb wave-based NDT systems for damage detection in composites using different types of sensors such as piezoelectric transducer, fiber Bragg grating, and laser Doppler vibrometer and Lamb wave generation methods such as piezoelectric transducers, Q-switched laser, or pulsed laser [4–7].

Ultrasonic propagation imaging (UPI) system is an effective technology for damage detection and offers fast inspection speeds and reliable results. The UPI system provides a better understanding of wave behavior in a spatial–temporal domain and shows how the wave field interacts with a structure [6,8]. UPI system consists of a laser scanning system, ultrasonic sensing system, and high-speed data acquisition

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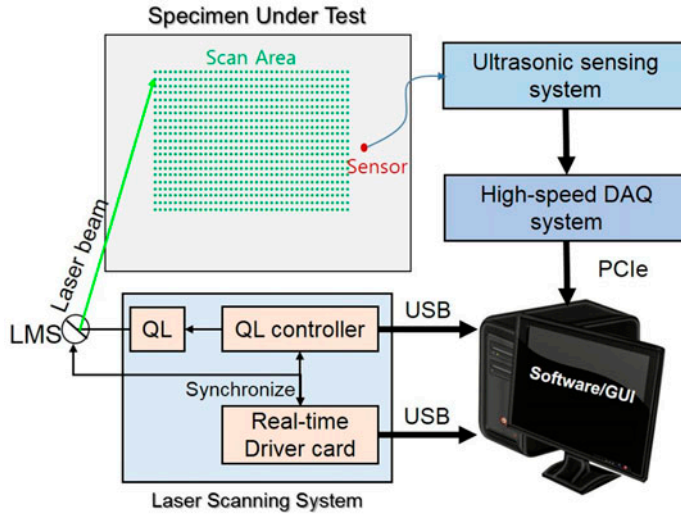


Figure 1. Overview of ultrasonic propagation imaging (UPI) system. LMS: Laser Mirror Scanner. QL: Q-Switched laser.

(DAQ) system as shown in Figure 1. The laser scanning utilizes a 532-nm Q-switched Nd:YAG laser and a laser mirror scanner (LMS) to perform high-speed pulse repetition of up to 20 kHz [9]. The LMS makes use of two orthogonal motorized mirrors, synchronous with the laser pulse beams, to perform two-dimensional inspection of the structure under testing. The sensing system can be used with various sensor types including contact and non-contact sensors. In this study, we used an integrated amplifier, piezoelectric transducer for sensing. A band-pass filter was applied to the ultrasonic signals before they were digitized. The high-speed DAQ consists of a field programmable gate array (FPGA)-based DAQ and a processing board installed on a computer PCIe slot. It is responsible for digitizing the ultrasonic signals and applying damage detection algorithms for fast processing and result generation [8,10,11].

Several damage detection algorithms are integrated into the FPGA-based UPI system to provide fast and enhanced inspection of the target structure. Previously, ultrasonic wave propagation imaging (UWPI) was implemented in FPGA and accelerated results up to eight times were achieved in post-processing result generation compared to CPU implementation [11]. In another study, multidimensional adjacent wave subtraction methods were integrated into an FPGA-based UPI system, and real-time results were generated with a 95.5% time reduction compared to a conventional CPU-based UPI system [9].

The ultrasonic energy mapping (UEM) technique is another promising method with the advantage of damage size assessment. UEM method involves two-dimensional Fourier transform and wavenumber filtering of full wavefield data. Previously, this algorithm was not integrated into the UPI system; therefore, more time and effort was required to convert the scan data and perform the post-processing. To use the UEM method in real-field applications, we need to locally process the algorithm in the UPI system. Integration of this technique for fast result generation in the UPI system first required implementation of the UEM in FPGA. This is a challenging task because multidimensional Fourier transform for a large number of data-sets cannot fit into a single; therefore, external memory to handle large data was needed [12].

In this study, we integrated UEM to the FPGA platform for the UPI system. Multidimensional Fourier transform was performed using a row-column decomposition method and memory access provided by utilizing two DDR3 memories. Since the energy of the laser ultrasonic waves dissipates when traveling, to maximize the inspected area in a one-time scan, the sensor should be installed inside the scan area. This causes a new issue not encountered by a previous studying using an UEM algorithm [13], in which the energy concentrated in the damage(s) is easily masked by the ‘source’ energy at the location of the sensor. This attributed to a technique in this paper of source removal, applied to highlight the damage in presence of high source energy [14,15]. The sensor location is identified automatically during scanning. An experiment was performed on a composite wing box specimen with impact damage to verify the effectiveness of the system. The results showed that we successfully detected the damage of a 20 mm × 20 mm size in the existence of a source inside the scan area. This result is generated in less than 5 s for a 240 × 240 mm² inspection area.

The structure of the article is as follows. First, the implementation and architecture of UEM in FPGA are described. Second, the details of the experiment on a composite wing box panel are discussed. Finally, results are presented to verify the reliability of UEM in FPGA-based UPI system. The article ends with a conclusion.

2. Implementation of UEM on FPGA platform

In this section, the theory of the ultrasonic energy mapping algorithm is first presented. Secondly, the source removal function is described and its integration in the UPI system is discussed. Finally, we showed implementation of the two-dimensional Fourier transform for UEM algorithm in FPGA.

2.1. Ultrasonic energy mapping

UEM takes the full wavefield data that are acquired using the UPI system and performs two-dimensional Fourier transform, wavenumber filtering, and point-by-point wave energy calculation of the complete scan area [13]. Mathematically, UEM is presented using Equations (1)–(4) as follows:

$$W(v_x, v_y, t) = F_{2D}\{w(x, y, t)\} \quad (1)$$

$$\tilde{W}(v_x, v_y, t) = W(v_x, v_y, t)\Phi(v_0, \sigma) \quad (2)$$

$$\tilde{w}(x, y, t) = F_{2D}^{-1}\{\tilde{W}(v_x, v_y, t)\} \quad (3)$$

$$E(x, y) = \int_{T_1}^{T_2} \frac{1}{2} \{\tilde{w}(x, y, t)\}^2 dt \quad (4)$$

where $w(x, y, t)$ denotes the full wavefield time domain data and its two-dimensional Fourier transformation is denoted as $W(v_x, v_y, t)$. A wavenumber filter is applied to $W(v_x, v_y, t)$, which gives us a narrow-band wavefield in the wavenumber domain $\tilde{W}(v_x, v_y, t)$. $\Phi(v_0, \sigma)$ is a binary wavenumber mask at a wavenumber central, v_0 and standard deviation, σ . F_{2D} and F_{2D}^{-1} are the operations of two-dimensional Fourier transform and two-dimensional inverse Fourier transform, respectively. An inverse Fourier

transform is then applied, which results in a filtered wavefield in the space–time domain $\tilde{w}(x, y, t)$. Finally, wave energy $E(x, y)$ is calculated at each grid point as shown in Equation (4).

2.2. Source removal algorithm

Source removal algorithm is a very effective for the inspection of large specimens, therefore it is worthy to discuss it in detail. To maximize the scan area, the ‘source,’ which is at the sensor location, is located inside the scan area. The energy of the source is easily masked by the energy trapped inside the damage area. In this case, an algorithm to reduce the energy of the source is needed to reveal the damage(s).

Source removal is performed in both spatial and temporal domains. In the spatial domain, source removal is applied by multiplying the original raw data with a matrix mask having the same size as the scan area, a 0 value inside radius r , with the center at the sensor location and 1 elsewhere, as shown in Equation (7). In the temporal domain, source removal is applied by selecting the time band T_1 to T_2 , where the waves flow across the damages, as shown in Equation (4). Since the wave energy at the source is more significant at the beginning part of the wave propagation, much of the energy at the source is suppressed and the wave energy at the damage is amplified.

$$M_{i,j} = \begin{cases} 0 & \text{if } (i - x_{SL}/r)^2 + (j - y_{SL}/r)^2 \leq 1 \\ 1 & \text{if } (i - x_{SL}/r)^2 + (j - y_{SL}/r)^2 > 1 \end{cases} \quad (5)$$

where r is the radius of source, x_{SL} and y_{SL} are the coordinates of the sensor location. The process of finding the coordinates of the sensor location can be performed autonomously as follows. Since the energy at the sensor location is the maximum in the scan area, the source is traced by finding the energy of each scan point and determining the location of the scan point with the maximum energy. If the ultrasonic signal captured, after an impinge point is $S_{x,y}$ and having a total number of samples N , the energy $E_{SL}(x, y)$ is calculated as follows:

$$E_{SL}(x, y) = \sum_{i=0}^{N-1} |S_{x,y}(i)|^2 \quad (6)$$

The source location is given by,

$$(x_{SL}, y_{SL}) = \arg \max_{(x,y)} \{E_{SL}(x, y)\} \quad (7)$$

2.3. Implementation of UEM in FPGA

In FPGA, multidimensional Fourier transform is performed on ultrasonic wavefield data using a row–column decomposition method. The row–column decomposition method is very suitable for parallel implementation of multidimensional Fourier transform and reduces the complex multiplications by a large factor [16]. A 2D discrete Fourier transform (DFT), $F(n_1, n_2)$ of a two-dimensional matrix $f(k_1, k_2)$ having $N \times M$ samples is given by,

$$F(n_1, n_2) = \sum_{k_2=0}^{M-1} \sum_{k_1=0}^{N-1} f(k_1, k_2) \cdot W_N^{k_1 n_1} \cdot W_M^{k_2 n_2} \quad (8)$$

where $W_N = e^{-j2\pi/N}$, $k_1 = 0, 1, 2, 3, \dots, N-1$, and $k_2 = 0, 1, 2, 3, \dots, M-1$. Two-dimensional DFT in Equation (8) is decomposed into multiple one-dimensional

DFTs using the row-column decomposition method. First, N row-wise one-dimensional DFT is performed and then M column-wise one-dimensional DFT, as shown in Equations (9) and (10).

$$\tilde{F}(n_1, k_2) = \sum_{k_1=0}^{N-1} f(k_1, k_2) \cdot W_N^{k_1 n_1} \quad (9)$$

$$F(n_1, n_2) = \sum_{k_2=0}^{M-1} \tilde{F}(n_1, k_2) \cdot W_M^{k_2 n_2} \quad (10)$$

Xilinx DFT/FFT IP core is used to compute one-dimensional Fourier transform in our firmware design. Each IP core has a different limitation for its available transformation lengths; therefore, we have used both IP cores to achieve maximum flexibility for the inspection area in our design. Memory access is provided by two DDR3 memories, where data are read and written multiple times from one memory to another, based on the row-column decomposition method. Figure 2 shows the complete flow chart for

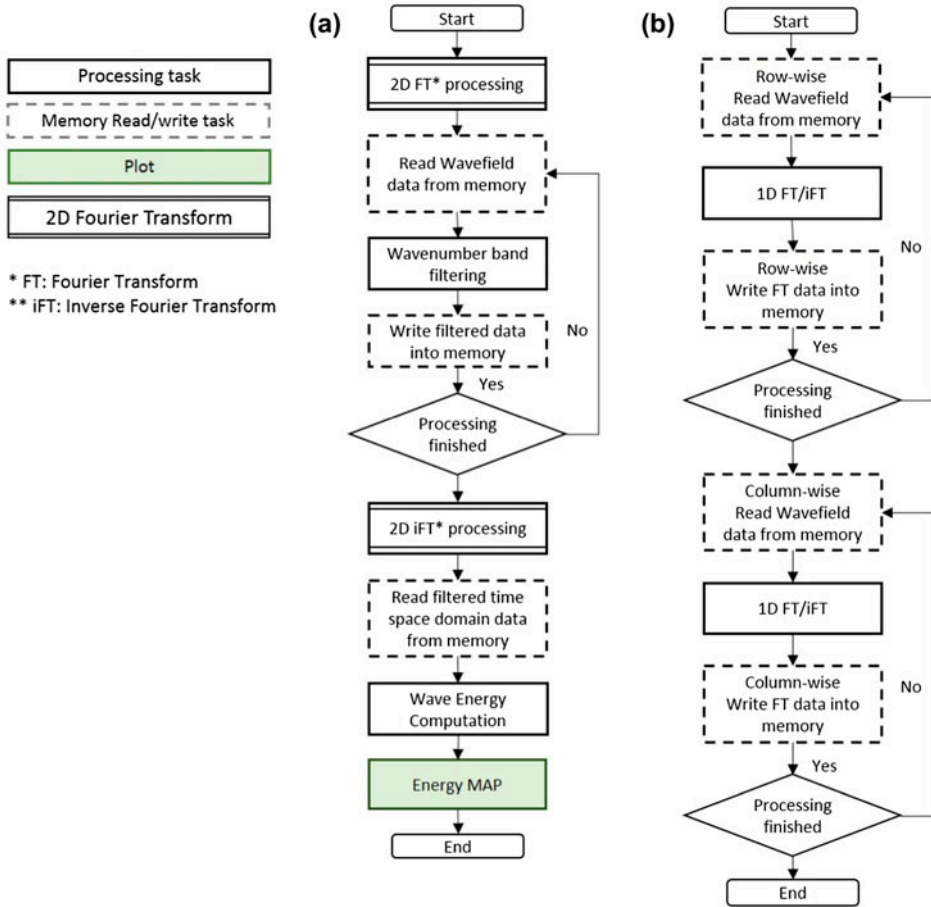


Figure 2. Flow chart for (a) UEM processing in FPGA, and (b) 2D Fourier transform process using row-column decomposition.

UEM implementation in FPGA. The overall UEM process in FPGA can be divided into read/write and processing tasks. Processing tasks are operations performed on the data received from the memory. Memory read/write tasks are used to fetch the data from the memory into the temporary buffer. Each of these tasks adds latency to the overall processing.

3. Experiment on composite wing box specimen

An experiment was performed on a composite wing box specimen with a PZT sensor attached to the back side of the specimen. The specimen has 2 mm thickness and $1125 \times 640 \text{ mm}^2$ area. The specimen also contains multiple stringers and a spar at the backside, where the sensor was attached. Figure 3 shows the front and rear view of the wing box structure providing the structure size, inspection area, sensor location, and defect position. The inspection was performed on the front size of the specimen from a 1.7-m stand-off distance to generate ultrasound waves using a Q-switched 532-nm Laser. The laser scanning was applied to a $240 \times 240 \text{ mm}^2$ area having a scan interval of 1 mm. The barely visible impact damage of size $21.4 \times 18.8 \text{ mm}^2$ is located at 80 mm distance from the left side of the scan area. The sensor was located inside the inspection area. The bandpass filter was applied between 100 and 200 kHz before digitizing the ultrasonic data. The data were acquired using an FPGA-based digitizer, setting the sampling frequency of 2.5 MHz and 512 samples per signal, which gives the total signal length of 205 μs . Figure 4 shows the experiment setup used to generate the UEM result.

4. Results

The UEM design is implemented on a Xilinx Virtix-6 XC6VLX240T FPGA device. To generate UEM results, first, we select the central wavenumber ν_0 which is critical for

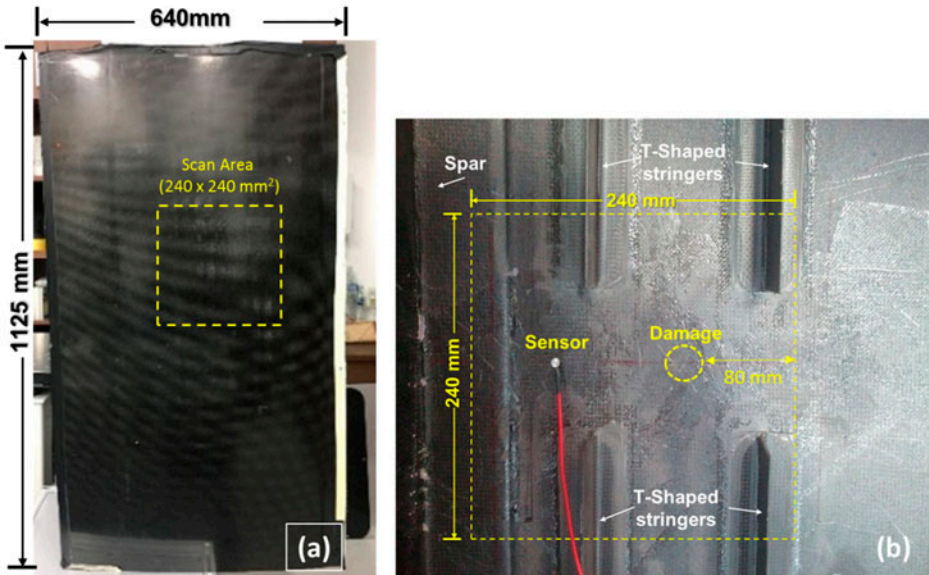


Figure 3. CFRP wing box structure. (a) Front view, and (b) Rear view.



Figure 4. Configuration of the experiment for CFRP wing box specimen inspection using PZT-UPI system.

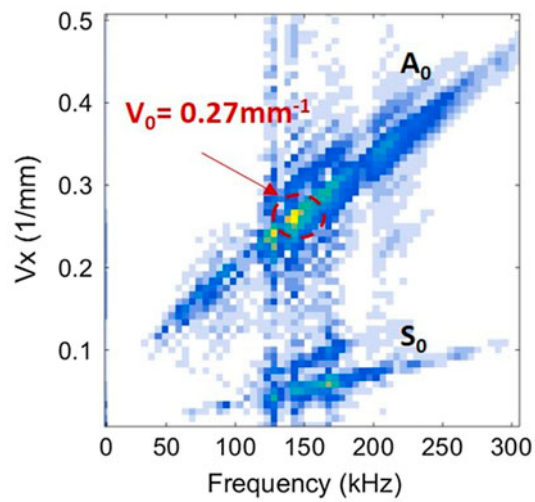


Figure 5. Wavenumber–frequency magnitude map.

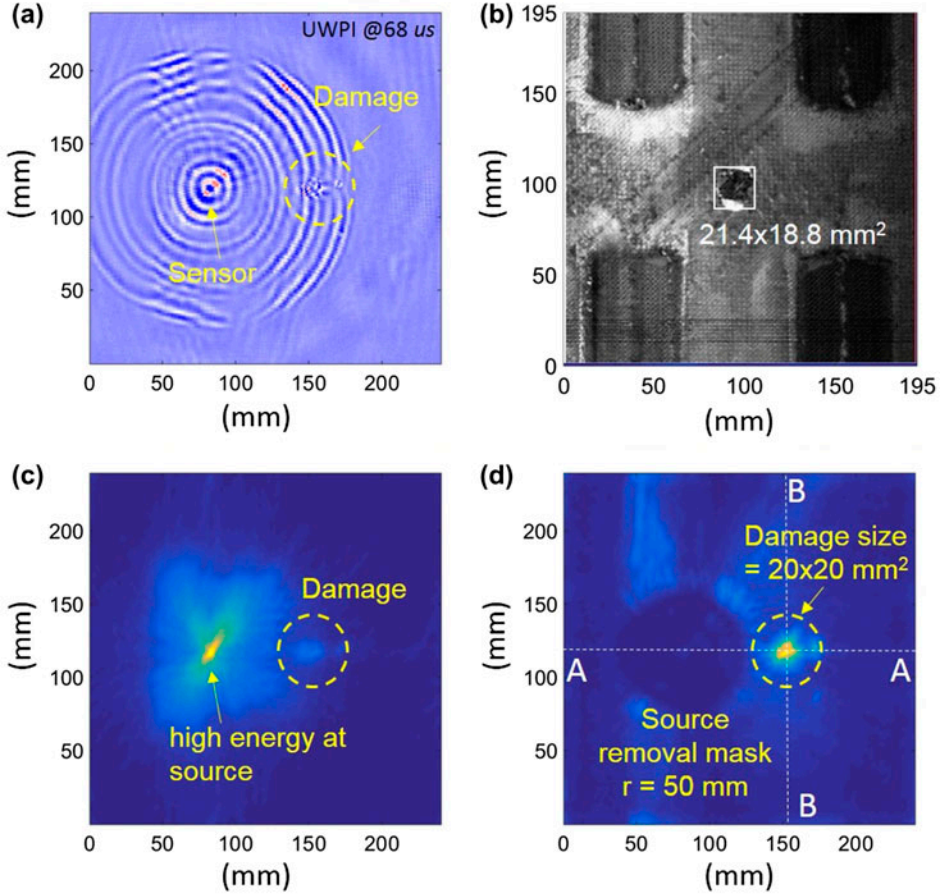


Figure 6. Inspection results from FPGA-based PZT-UPI system. (a) Freeze frame from ultrasonic wave propagation imaging (UWPI) result at $68 \mu\text{s}$, (b) Ultrasonic C-scan image, (c) FPGA-based UEM without applying source removal, and (d) FPGA-based UEM with source removal.

UEM wavenumber filtering. It is obtained from the wavenumber–frequency curve as shown in Figure 5. The wavenumber–frequency magnitude map shows two wave modes: asymmetrical mode (A_0) and symmetrical mode (S_0). The central wavenumber is selected on A_0 mode because the A_0 mode is generally more sensitive to damage since it has a shorter wavenumber. A central wavenumber of 0.27 mm^{-1} is applied to generate the UEM plot. Figure 6 shows the inspection results on a targeted composite structure. The ultrasonic wave propagation imaging (UWPI) freeze frame [8] is captured at $68 \mu\text{s}$, showing the scattering waves at the damage location. UWPI result is very effective to locate the damage but we cannot evaluate the size of the damage using this result as shown in Figure 6(a). C-scan image is taken using commercial device to find the actual size of the impact damage which gives us the defect size of $21.4 \times 18.8 \text{ mm}^2$ as shown in Figure 6(b). Figure 6(c) and (d) show the contrast between two results of FPGA-based UEM that are generated with and without a source removal technique. The result showed that impact damage of a $20 \times 20 \text{ mm}^2$ size was more visible when the source removal mask was applied to the scan area. The damage

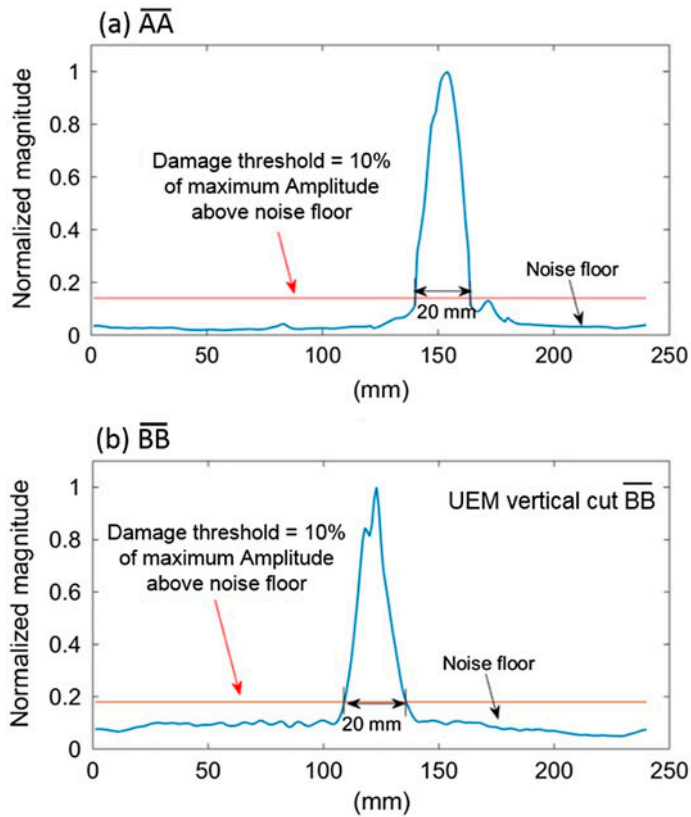


Figure 7. Damage sizing and threshold. (a) Horizontal cut, and (b) Vertical cut.

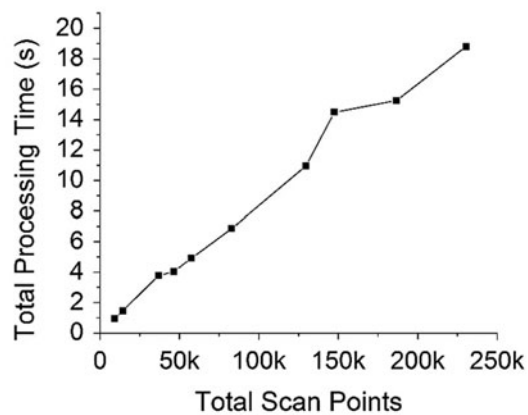


Figure 8. Performance graph of FPGA-based UEM.

sizing is determined using damage threshold method. Damage threshold is the percentage of maximum amplitude above the noise floor [14]. To determine damage and its size, the vertical and horizontal lines are obtained corresponding to the y -coordinate and x -coordinate of the maximum amplitude point on UEM map. The 10% damage threshold was applied to both horizontal and vertical lines as shown in Figure 7. The measured damage size is 20 mm long and 20 mm wide with an error margin of $\pm 7\%$.

A performance graph is shown in Figure 8, which presents the processing time to generate the damage visualization results based on the UEM method using FPGA. These results are obtained for various inspection sizes and show that it takes less than 5 s to inspect a scan area of 240×240 points.

5. Conclusions

In this paper, the implementation of the UEM method in FPGA was presented, and a source removal technique was suggested for composite structures. UEM required a two-dimensional Fourier transform operation to be computed in FPGA, which was performed using a row-column decomposition method. A source removal function was also incorporated with the UEM technique. Using a source removal technique, the system can visualize the damage in the presence of a high-energy source. During laser scanning for Lamb wave generation on the specimen, source location was identified by calculating the energy of each impinging point and finding the maximum energy point. The source removal function was then applied by first multiplying a mask at the sensor location in the spatial domain and then filtering in the temporal domain.

An experiment was performed on a composite wing box panel made of carbon fiber-reinforced plastic (CFRP) to obtain the laser ultrasonic wavefield data using the FPGA-based UPI system. Although the sensor was located inside the scan area, using the source removal method, the UEM result was able to detect a $20 \times 20 \text{ mm}^2$ damage size. The results showed that the wavefield time domain data of a total 57,600 points was processed in less than 5 s to generate the UEM result. These results have proved that the UEM method can be successfully implemented in FPGA and integrated with the UPI system for real composite damage evaluation.

Disclosure statement

No potential conflict of interest was reported by the authors.

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